

NATIONAL OPEN UNIVERSITY OF NIGERIA
University Village Plot 91, Cadastral Zone, Nnamdi Azikiwe Expressway, Jabi, Abuja
FACULTY OF SCIENCES
DEPARTMENT OF MATHEMATICS
2023_1 POP EXAMINATION...

Course Code: MTH301
Course Title: Functional Analysis I
Credit Unit: 3
Time Allowed: 3 Hours
Total: 70 Marks
Instruction: Answer Question One (1) and Any Other 3 Questions

- (1) (a) Define (i) a complete metric space **(3 Marks)**
(ii) a first category set **(3 Marks)**
(iii) a Countable set **(3 Marks)**
(iv) an Open ball. **(3 Marks)**
- (b) (i) Let X be a complete metric space and $\{O_n\}$ be a countable collection of dense open subsets of X . Prove that the union $\bigcup O_n$ is not empty. **(4 Marks)**
(ii) Prove that in $\mathbb{R}^n$, every family of disjoint non-empty open set is countable. **(3 Marks)**
- (c) (i) Prove that in $\mathbb{R}^n$, the union of arbitrary collection of open sets is open. **(3 Marks)**
(ii) Prove that the finite intersection of a collection of open sets is open. **(3 Marks)**
- (2) (a) Define the followings
(i) Pseudometric **(4.5 Marks)**
(ii) Distance between two vectors **(4.5 Marks)**
- (b) Explain the concept of an ordered field. **(6 Marks)**
- (3) (a) Define the followings:
(i) a function **(3 Marks)**
(ii) real-valued function **(3 Marks)**
(iii) vector-valued function. **(3 Marks)**
- (b) Let A and B be metric spaces. Prove that $f: A \rightarrow B$ is continuous if and only if $f^{-1}(V)$ is an open set in A whenever V is open in B . **(6 Marks)**
4. (a) Define the followings
(i) neighborhood of a point of a metric space **(3 Marks)**
(ii) connected topological space **(3 Marks)**
(iii) open cover for a subset S of a metric space (X, d) **(3 Marks)**
(iv) compact subset of a metric space. **(3 Marks)**
- (b) Write out two examples of compact subsets of a metric space (X, d) . **(3 Marks)**

(5) Write out two examples of the neighborhood of the point p in a metric space.

(6 Marks)

(b) Let (K, d) be a compact metric space. Prove that every sequence in K has a convergent subsequence.

(9 Marks)

(6) For any vector $X \in \mathfrak{R}^n$, state the theorems concerning

(i) the norm, for all $x, y \in X, \alpha \in \mathfrak{R}$.

(5 Marks)

(ii) the metric d , for all $x, y, z \in X, \alpha \in \mathfrak{R}^n$.

(5 Marks)

(iii) Inner product, for all x, y, y_1 , and $y_2 \in X$ and $\alpha \in \mathfrak{R}$.

(5 Marks)